

```
[> # e)
```

```
> ec1:=diff(x[1](t),t)=-2*x[2](t)
```

$$ec1 := \frac{d}{dt} x_1(t) = -2 x_2(t) \quad (50)$$

```
> ec2:=diff(x[2](t),t)=x[1](t)-2*x[2](t)
```

$$ec2 := \frac{d}{dt} x_2(t) = x_1(t) - 2 x_2(t) \quad (51)$$

```
> ec3:=diff(x[3](t),t)=-2*x[3](t)
```

$$ec3 := \frac{d}{dt} x_3(t) = -2 x_3(t) \quad (52)$$

```
> sist:=ec1,ec2,ec3;
```

$$sist := \frac{d}{dt} x_1(t) = -2 x_2(t), \frac{d}{dt} x_2(t) = x_1(t) - 2 x_2(t), \frac{d}{dt} x_3(t) = -2 x_3(t) \quad (53)$$

```
> sol:=dsolve({sist},{x[1],x[2],x[3]});
```

$$sol := \left\{ x_1(t) = e^{-t} (\sin(t) c_1 + \cos(t) c_2), x_2(t) = \frac{e^{-t} (\sin(t) c_1 + \sin(t) c_2 - \cos(t) c_1 + \cos(t) c_2)}{2}, x_3(t) = c_3 e^{-2t} \right\} \quad (54)$$

```
> rhs(sol[1]);
```

$$e^{-t} (\sin(t) c_1 + \cos(t) c_2) \quad (55)$$

```
> limit(rhs(sol[1]),t=infinity);
```

$$0 \quad (56)$$

```
> limit(rhs(sol[2]),t=infinity);
```

$$0 \quad (57)$$

```
> limit(rhs(sol[3]),t=infinity);
```

$$0 \quad (58)$$

```
> # Exercise 2
```

```
> restart;
```

```
> with(LinearAlgebra)
```

[&x, Add, Adjoint, BackwardSubstitute, BandMatrix, Basis, BezoutMatrix, BidiagonalForm, BilinearForm, CARE, CharacteristicMatrix, CharacteristicPolynomial, Column, ColumnDimension, ColumnOperation, ColumnSpace, CompanionMatrix, CompressedSparseForm, ConditionNumber, ConstantMatrix, ConstantVector, Copy, CreatePermutation, CrossProduct, DARE, DeleteColumn, DeleteRow, Determinant, Diagonal, DiagonalMatrix, Dimension, Dimensions, DotProduct, EigenConditionNumbers, Eigenvalues, Eigenvectors, Equal, ForwardSubstitute, FrobeniusForm, FromCompressedSparseForm, (59)

FromSplitForm, GaussianElimination, GenerateEquations, GenerateMatrix, Generic, GetResultDataType, GetResultShape, GivensRotationMatrix, GramSchmidt, HankelMatrix, HermiteForm, HermitianTranspose, HessenbergForm, HilbertMatrix, HouseholderMatrix, IdentityMatrix, IntersectionBasis, IsDefinite, IsOrthogonal, IsSimilar, IsUnitary, JordanBlockMatrix, JordanForm, KroneckerProduct, LA_Main, LUDecomposition, LeastSquares, LinearSolve, LyapunovSolve, Map, Map2, MatrixAdd, MatrixExponential, MatrixFunction, MatrixInverse, MatrixMatrixMultiply, MatrixNorm, MatrixPower, MatrixScalarMultiply, MatrixVectorMultiply, MinimalPolynomial, Minor, Modular, Multiply, NoUserValue, Norm, Normalize, NullSpace, OuterProductMatrix, Permanent, Pivot, PopovForm, ProjectionMatrix, QRDecomposition, RandomMatrix, RandomVector, Rank, RationalCanonicalForm, ReducedRowEchelonForm, Row, RowDimension, RowOperation, RowSpace, ScalarMatrix, ScalarMultiply, ScalarVector, SchurForm, SingularValues, SmithForm, SplitForm, StronglyConnectedBlocks, SubMatrix, SubVector, SumBasis, SylvesterMatrix, SylvesterSolve, ToeplitzMatrix, Trace, Transpose, TridiagonalForm, UnitVector, VandermondeMatrix, VectorAdd, VectorAngle, VectorMatrixMultiply, VectorNorm, VectorScalarMultiply, ZeroMatrix, ZeroVector, Zip]

> D1:=LinearAlgebra:-DiagonalMatrix([2,2,-1,0]);

$$D1 := \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(60)

> P:=LinearAlgebra:-RandomMatrix(4,generator=-1..1);

$$P := \begin{bmatrix} -1 & 0 & 1 & 1 \\ -1 & 0 & -1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

(61)

> A:=simplify(P.D1.P^(-1));

$$A := \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & 0 & -2 \\ \frac{3}{2} & -\frac{3}{2} & 0 & -2 \\ -\frac{3}{2} & -\frac{1}{2} & 2 & -2 \\ -\frac{3}{2} & \frac{3}{2} & 0 & 2 \end{bmatrix} \quad (62)$$

> eigenvals(A)

$$\text{eigenvals} \left(\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & 0 & -2 \\ \frac{3}{2} & -\frac{3}{2} & 0 & -2 \\ -\frac{3}{2} & -\frac{1}{2} & 2 & -2 \\ -\frac{3}{2} & \frac{3}{2} & 0 & 2 \end{bmatrix} \right) \quad (63)$$

> eigenvalues(A)

$$\text{eigenvalues} \left(\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & 0 & -2 \\ \frac{3}{2} & -\frac{3}{2} & 0 & -2 \\ -\frac{3}{2} & -\frac{1}{2} & 2 & -2 \\ -\frac{3}{2} & \frac{3}{2} & 0 & 2 \end{bmatrix} \right) \quad (64)$$

> Eigenvalues(A);

(65)

$$\begin{bmatrix} 0 \\ -1 \\ 2 \\ 2 \end{bmatrix}$$

(65)

> Eigenvectors(A)

$$\begin{bmatrix} 0 \\ -1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 & 1 & -1 & 0 \\ 1 & -1 & -1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

(66)

> # b)

> Determinant(A)

0

(67)

> CharacteristicPolynomial(A,lambda)

$$\lambda^4 - 3\lambda^3 + 4\lambda$$

(68)

> solve(lambda^4 - 3*lambda^3 + 4*lambda=0,lambda);

0, -1, 2, 2

(69)

> Eigenvectors(A);

$$\begin{bmatrix} 0 \\ 2 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & -1 & 1 \\ 1 & 0 & -1 & -1 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

(70)

> JordanForm(A);

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

(71)

> MatrixInverse(A);

Error. (in MatrixInverse) singular matrix

> # METHOD GAUSS-JORDAN: zeroes both under and above main diagonal
- used for big stuff

> # Exercise 3